

1. Difference Between Fit() and Fit_transform():

1- Fit (): Learns model parameters (mean and standard deviation for normalization) from the training data but does not apply the transformation.

2- Fit_transform(): Combines fit() and transform() into a single step, which is more efficient for processing training data. Fit() to prepare transformers (like scaler) on training data, then transform() for testing data to prevent data leakage.

2. Data Leakage :

Data leakage refers to a situation where information that should not be accessible to the ML model during training becomes part of the training dataset. This misleading information artificially impacts the model's performance metrics during training and evaluation but causes it to fail when exposed to new, unseen data.

Types of Data Leakage :

- Target Leakage.
- Temporal Cut-off.
- Feature Leakage.
- Train Test Contamination.

Train-Test Contamination : Happens when information from the test or validation set "leaks" into the training process, often due to improper data handling before the data is split.